2 **(a)** State Newton's second law of motion.

.....

.....

.....[1]

(b) A ball of mass 150 g travelling horizontally at a speed of 20 m s^{-1} collides with a vertical wall. Fig. 2.1 shows how the force of impact on the wall, F, varies with time t.

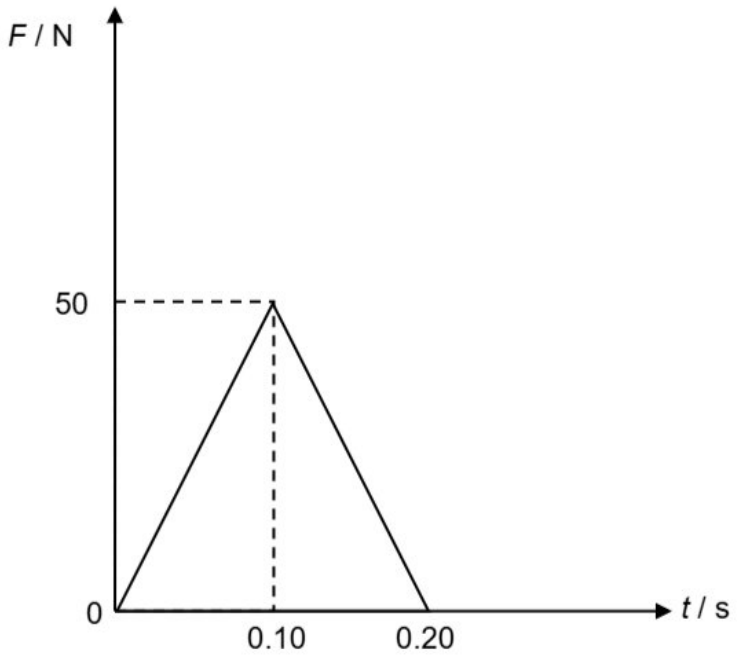


Fig. 2.1

(i) Determine the change in momentum of the ball.

change in momentum = ... N s [2]

(ii) Hence determine the final speed of the ball.

final speed = ... m s^{-1} [1]

(iii) Balls which are identical to the one above are thrown at the wall at the rate of 30 balls per minute. Determine the average impact force on the wall.

average impact force = ... N [2]

(iv) A ball which is equal in mass as the one above is made of a harder material. It also collides with the wall at the same speed as before. Draw on Fig. 2.1 a second graph to show how the impact force on the wall varies with time when this ball collides with the wall. [1]

[Total: 7]